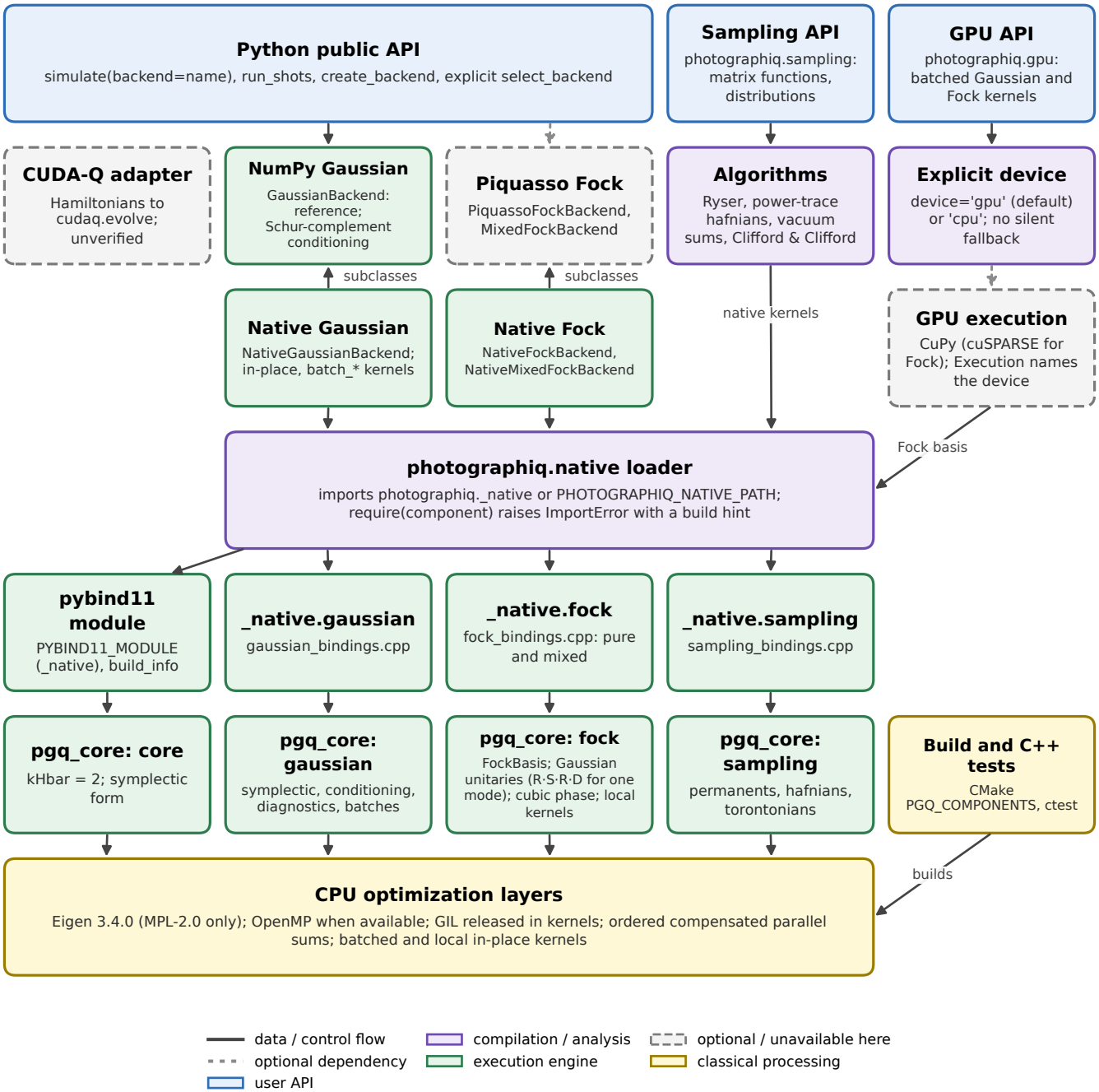C Simulation architecture